

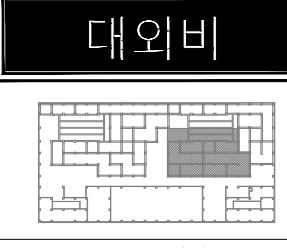
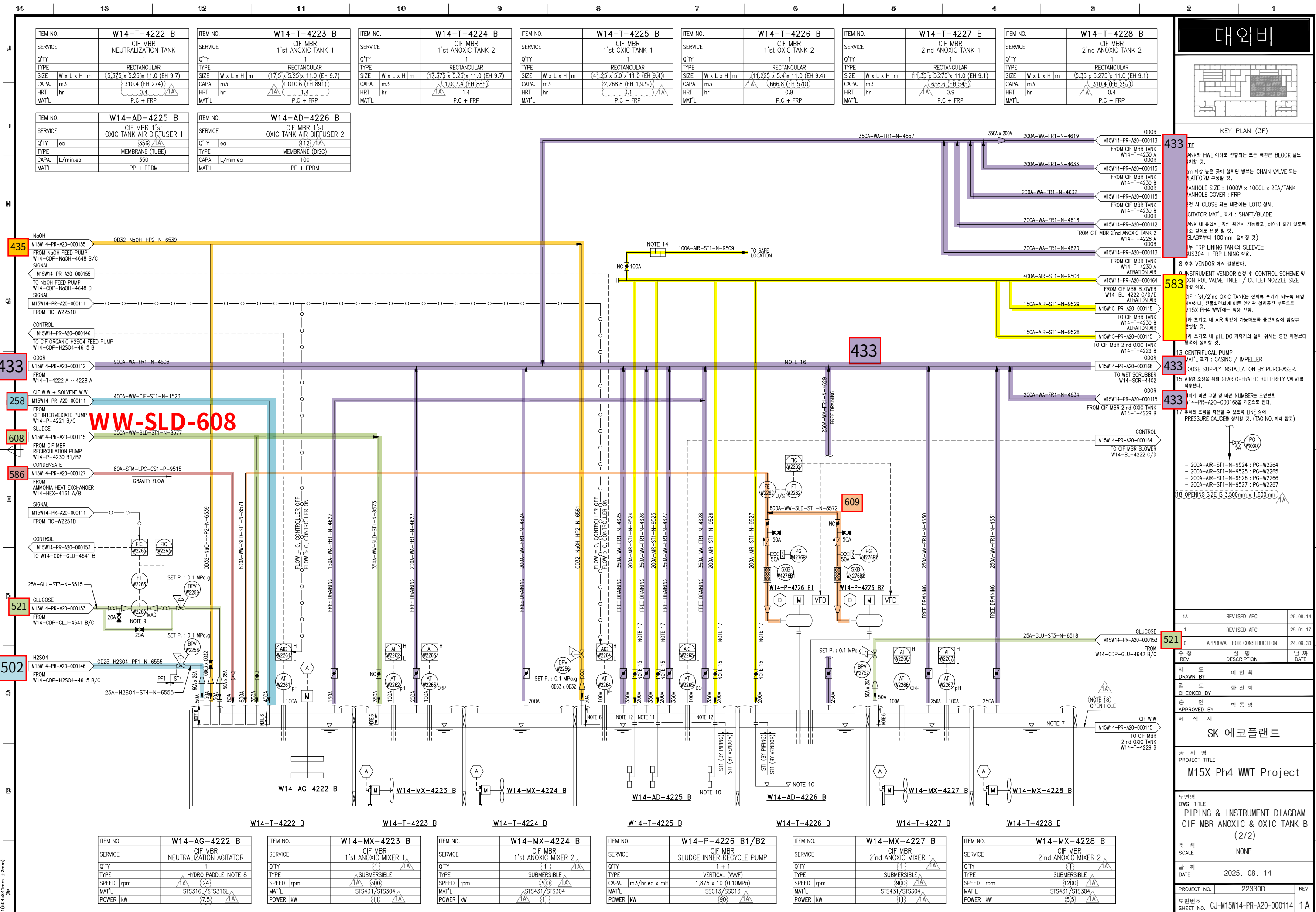
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

WW-SLD-608

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	608	3FA	350A-WW-SLD-ST1-N-8577-01	1 OF 2	1	CJ-M15W14-PR-A20-000115	2A	
2	608	3FA	350A-WW-SLD-ST1-N-8577-02	2 OF 2	0	CJ-M15W14-PR-A20-000115	2A	
3	608	3FA	350A-WW-SLD-ST1-N-8578-01	1 OF 1	1	CJ-M15W14-PR-A20-000115	2A	
4	608	4FA	350A-WW-SLD-ST1-N-8573-01	1 OF 1	0	CJ-M15W14-PR-A20-000114	1A	
5	608	4FA	350A-WW-SLD-ST1-N-8577-01	1 OF 1	0	CJ-M15W14-PR-A20-000115	2A	



대외비

KEY PLAN (3F)

433

1. ANOXIC H/W 이하로 연결되는 모든 배관은 BLOCK 밸브 치환할 것.

2. m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.

3. ANHOLE SIZE : 1000W x 1000L x 2EA/TANK ANHOLE COVER : FRP

4. 전 시 CLOSE 되는 배관에는 LOTO 설치.

5. GITATOR MAT'L 표기 : SHAFT/BLADE

6. ANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 소 길이로 변경 할 것.

7. SLAB로부터 100mm 떨어진 것.

8. FRP LINING TANK에 SLEEVE는 SUS304 + FRP LINING 적용.

9. B.주후 VENDOR에서 결정한다.

10. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 결정 예정.

11. C/F 1st/2nd OXIC TANK는 전회류 포기가 되도록 배설해야하나, 건물외벽에 따른 순간 설치공간 부족으로 M15X PH4 WWT에는 적용 안함.

12. 차 호기조 내 AIR 확인이 가능하도록 중간지점에 점검구 구성할 것.

13. 차 호기조 내 pH, DO 계측기의 설치 위치는 중간 지점보다 양쪽에 설치할 것.

14. 13. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER

15. DOOSE SUPPLY INSTALLATION BY PURCHASER.

16. 15. AIR용 조정을 위해 GEAR OPERATED BUTTERFLY VALVE를 적용한다.

17. 433 배관 구성 및 배관 NUMBER는 도면번호 W14-PR-A20-000168을 기준으로 한다.

18. 17. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치할 것. (TAG NO. 아래 참조)

19. 18. OPENING SIZE IS 3,500mm x 1,600mm

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	이인학
검	토	한진희
CHECKED BY		
승인	박동영	
APPROVED BY		
제	작	사
SK 에코플랜트		
공	사	명
PROJECT TITLE	M15X Ph4 WWT Project	
도면명	PIPING & INSTRUMENT DIAGRAM	
DWG. TITLE	C/F MBR ANOXIC & OXIC TANK B	
	(2/2)	
속	척	
SCALE	NONE	
날	짜	
DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호	CJ-M15W14-PR-A20-000114	1A

ITEM NO.	W14-AD-4229 B
SERVICE	CIF MBR 2'nd OXIC TANK AIR DIFFUSER
Q'TY	110
TYPE	MEMBRANE (DISC)
CAPA.	L/min.ea
MAT'L	PP + EPDM

ITEM NO.	W14-T-4229 B
SERVICE	CIF MBR 2'nd OXIC TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	min
MAT'L	R.C + FRP

ITEM NO.	W14-T-4230 B
SERVICE	CIF MBR TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	hr
MAT'L	P.C + FRP

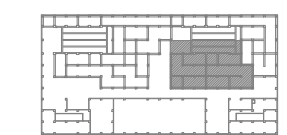
ITEM NO.	W14-T-4232 B
SERVICE	CIF MBR CLEANING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	min
MAT'L	C.S + FRP

ITEM NO.	W14-AD-4230 B
SERVICE	CIF MBR AIR DIFFUSER
Q'TY	ea
TYPE	MEMBRANE (DISC)
CAPA.	L/min.ea
MAT'L	PP + EPDM

ITEM NO.	W14-EJ-4231 B
SERVICE	CIF MBR EJECTOR UNIT
Q'TY	ea
TYPE	BY VENDOR
CAPA.	N/s.ea
MAT'L	BY VENDOR

ITEM NO.	W14-MBR-4230 B
SERVICE	CIF MBR MEMBRANE
CAPA.	m3/d
	15,000

대외비



KEY PLAN (3F)

- NOTE
1. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 2. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 3. MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP
 4. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
 5. CENTRIFUGAL PUMP
MAT'L 표기 : CASING / IMPELLER
 6. 4부 FRP LINING TANK SLEEVE는 SUS304 + FRP LINING 적용.
 7. PACKAGE VENDOR 선정 후 4부 SYSTEM (공회 EQUIPMENT SPEC., LINE SIZE, MBR Nos., INSTRUMENT CONTROL SCHEME 등) 확정 예정.
 8. 안전포기.
 9. 차단포기.
 10. 용압계 설치.
DESIGN RANGE : -1~1 BAR

INTERLOCK NO. 1-4231 B (W14-P-4231 B1/B2 AUTO STOP)
RENOTE STATUS에 W14-T-4229 B LEVEL (LI-W2321) LOW LOW STOP.

12. OPERATION TIME CHART.

PROCESS	SERVICE	B/W READY	B/W	RESTART
TIME(sec)	24hr	2	30	30
XV-2331A	OPEN	-	-	-
XV-2331B	-	-	OPEN	-
XV-2321	-	OPEN	-	OPEN
B/W PUMP	-	-	RUN	-

4. 나머지 ON-OFF VALVE들도 동일하게 적용한다.
13. CIF MBR TANK CRANE를 PACKAGE VENDOR에게 공급한다.
14. DELETED.
15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 확정 예정.
16. MANHOLE 2개소 설치하여야 한다.
17. SUPPLIED BY MBR PACKAGE VENDOR.
18. HIGH POINT에 SIPHON BREAKER 설치할 것.
(CHECK VALVE의 경우 DUAL PLATE TYPE으로 적용)
19. 개개의 세정용 CASSETTE CONNECTION 용도.
MBR CLEANING TANK 인근에 설치할 것.
20. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치할 것. (TAG NO. 아래 참조)

- 150A-AIR-ST1-N-9528 : PG-W2381
21. CIF MBR CLEANING TANK SHALL BE INSTALLED COVER WITH DUCT. OVERFLOW LINE(80A) TO DRAIN HEADER.
22. 하도개 설치를 위한 FUTURE CONNECTIONS.
23. PORT SIZE 350A. (TEE + 단관 + CAP 마감 : 단관 끝에는 W14-P-4231 PUMP의 CASING 일단 보다 높게 제작)
24. AIR RELEASE VALVE
25. SET POINT : 6BAR ~> 3 ~1 BAR
26. SIZE IS 3,500mm x 1,600mm

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정	실명	날짜
REV.	DESCRIPTION	DATE

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
CIF MBR DISCHARGE & CIRCULATION PUMP B (2/2)

프로젝트명
PROJECT TITLE
M15X Ph4 WWT Project

도면번호
DWG. NO.
22330D

날짜
DATE
2025. 08. 14

도면번호
DWG. NO.
22330D

날짜
DATE
2025. 08. 14

도면번호
DWG. NO.
22330D

날짜
DATE
2025. 08. 14

도면번호
DWG. NO.
22330D

ITEM NO.	W14-P-4231 B1/B2
SERVICE	CIF MBR SUCTION PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VWF)
CAPA.	m3/hr x mH
MAT'L	SSC13/SSC13
POWER kW	(132)

ITEM NO.	W14-P-4230 B1/B2
SERVICE	CIF MBR RECIRCULATION PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL
CAPA.	m3/hr x mH
MAT'L	SSC13/SSC13
POWER kW	(90)

DRAIN
M15W13-PR-A20-000084
TO UDD FOR DRAIN HEADER

150A-D-ST1-N-9514

400A x 300A

400A-WW-SLD-ST1-N-8576

400A-WW-SLD-ST1-N-8577

400A-WW-SLD-ST1-N-8578

400A-WW-SLD-ST1-N-8579

50A-WW-SLD-ST1-N-8579

50A-WW-SLD-ST1-N-9515

50A-WW-SLD-ST1-N-9515

50A-WW-SLD-ST1-N-9515

50A-WW-SLD-ST1-N-9515

W14-T-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-AD-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-T-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-AD-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-T-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-AD-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-T-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-AD-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

ITEM NO.	W14-AD-4229 B
SERVICE	CIF MBR 2'nd OXIC TANK AIR DIFFUSER
Q'TY	110
TYPE	MEMBRANE (DISC)
CAPA.	L/min.ea
MAT'L	PP + EPDM

ITEM NO.	W14-T-4229 B
SERVICE	CIF MBR 2'nd OXIC TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	min
MAT'L	R.C + FRP

ITEM NO.	W14-T-4230 B
SERVICE	CIF MBR TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	hr
MAT'L	P.C + FRP

ITEM NO.	W14-T-4232 B
SERVICE	CIF MBR CLEANING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m
CAPA.	m3
HRT	min
MAT'L	C.S + FRP

ITEM NO.	W14-AD-4230 B
SERVICE	CIF MBR AIR DIFFUSER
Q'TY	ea
TYPE	MEMBRANE (DISC)
CAPA.	L/min.ea
MAT'L	PP + EPDM

ITEM NO.	W14-EJ-4231 B
SERVICE	CIF MBR EJECTOR UNIT
Q'TY	ea
TYPE	BY VENDOR
CAPA.	N/s.ea
MAT'L	BY VENDOR

ITEM NO.	W14-MBR-4230 B
SERVICE	CIF MBR MEMBRANE
CAPA.	m3/d
	15,000

ITEM NO.	W14-P-4231 B1/B2
SERVICE	CIF MBR SUCTION PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VWF)
CAPA.	m3/hr x mH
MAT'L	SSC13/SSC13
POWER kW	(132)

ITEM NO.	W14-P-4230 B1/B2
SERVICE	CIF MBR RECIRCULATION PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL
CAPA.	m3/hr x mH
MAT'L	SSC13/SSC13
POWER kW	(90)

DRAIN
M15W13-PR-A20-000084
TO UDD FOR DRAIN HEADER

150A-D-ST1-N-9514

400A x 300A

400A-WW-SLD-ST1-N-8576

400A-WW-SLD-ST1-N-8577

400A-WW-SLD-ST1-N-8578

400A-WW-SLD-ST1-N-8579

50A-WW-SLD-ST1-N-8579

50A-WW-SLD-ST1-N-9515

50A-WW-SLD-ST1-N-9515

50A-WW-SLD-ST1-N-9515

50A-WW-SLD-ST1-N-9515

W14-T-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-AD-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-T-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-AD-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-T-4229 B
NOTE 16

W14-MBR-4230 B
W14-T-4230 B

W14-AD-4230 B

W14-T-4232 B
NOTE 17

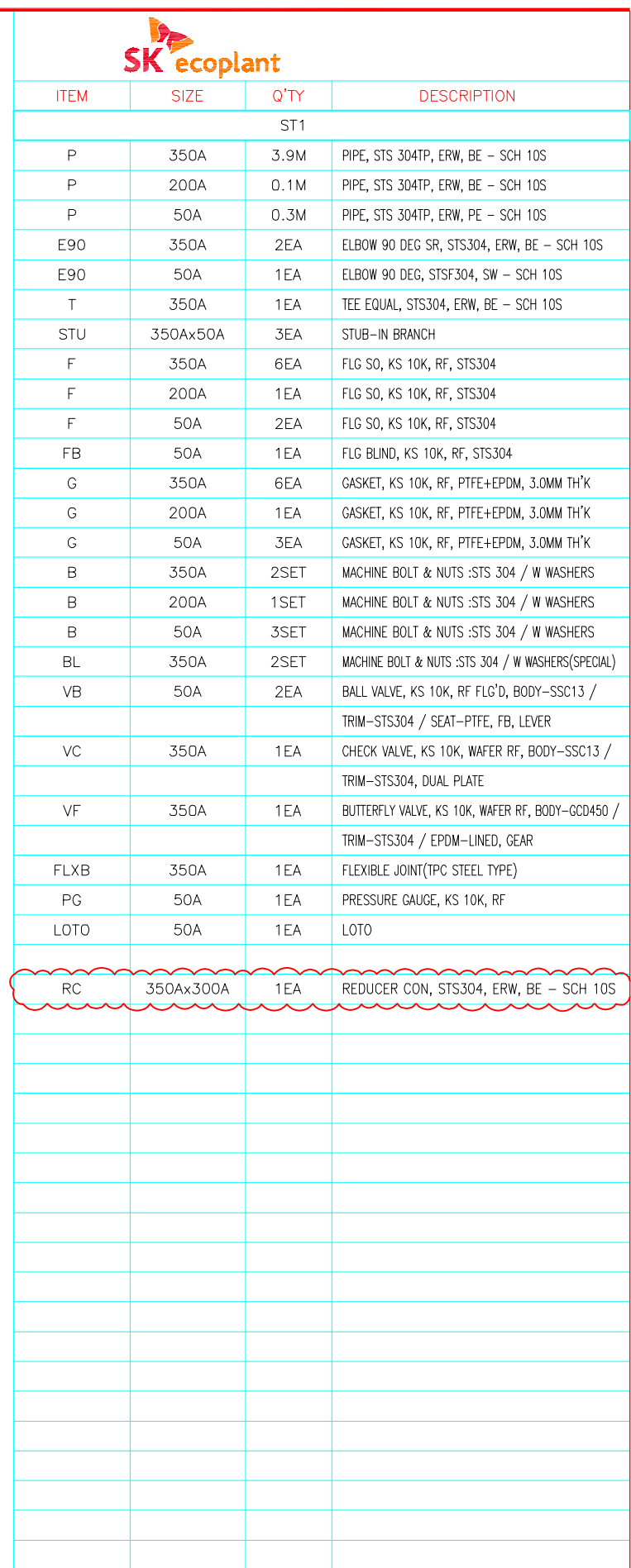
W14-P-4230 B1
W14-P-4230 B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-P-4230 B1/B2

W14-AD-4229 B
NOTE 16



CLIENT  1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	3.1	60	UNIT	3FA								M15X Ph4 Project	
			OPERATING	2.0	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-030001-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	WW-SLD	JOB NO.
					REF. P. & ID	CJ-M15W14-PR-A20-000115		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	350A-WW-SLD-ST1-N-8577	SH'T NO.	REV.
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			01 / 02	1

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CLIENT

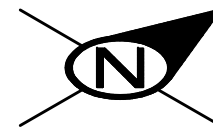


2

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(I-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	3.1	60	UNIT 3FA							M15X Ph4 Project	
			OPERATING	2.0	AMB	REF. PLAN DWG.						COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-030006-D						WW-SLD	22330D	
						REF. P & ID						LINE NO.	SH'T NO.	REV.
					CJ-M15W14-PR-A20-000115							350A-WW-SLD-ST1-N-8577	02 / 02	0
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

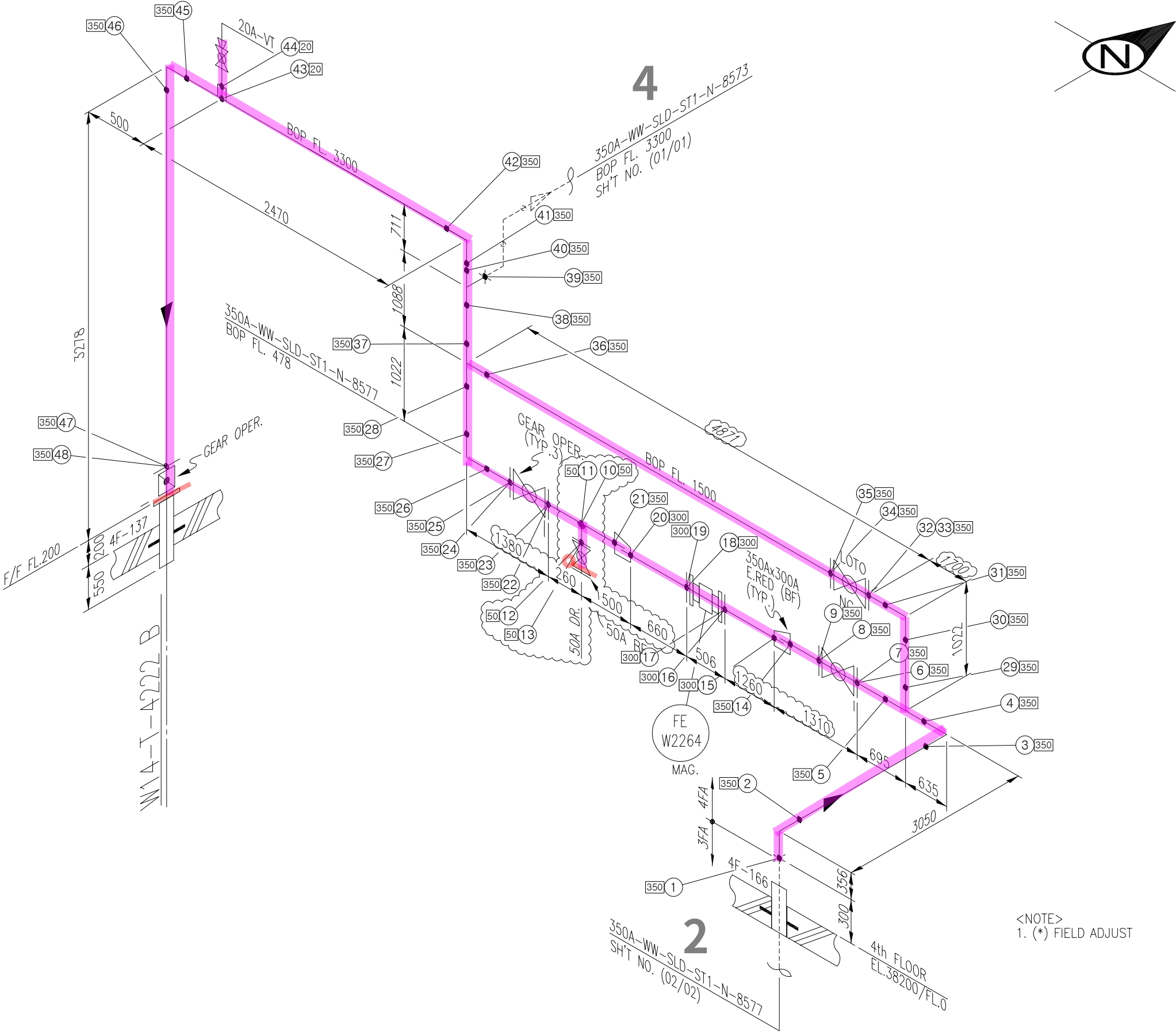


CLIENT  3



3

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



SK ecoplant			
ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	350A	15.5M	PIPE, STS 304TP, ERW, BE – SCH 10S
P	300A	2.8M	PIPE, STS 304TP, ERW, BE – SCH 10S
P	50A	0.1M	PIPE, STS 304TP, ERW, BE – SCH 10S
NP1	20A	1EA	NIPPLE, STS 304TP, ERW, TOE – SCH 40S
E90	350A	6EA	ELBOW 90 DEG SR, STS304, ERW, BE – SCH 10S
T	350A	3EA	TEE EQUAL, STS304, ERW, BE – SCH 10S
RE	350Ax300A	2EA	REDUCER ECC, STS304, ERW, BE – SCH 10S
STU	350Ax50A	1EA	STUB-IN BRANCH
HF	20A	1EA	COUPLING HALF, STSF304, SW – SCH 10S
PLG	20A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	1EA	BALL VALVE, KS 10K, SCR'D, BODY–STSF304 / TRIM–STS304 / SEAT–PTFE, FB, LEVER
VB	350A	3EA	BALL VALVE, KS 10K, RF FLG'D, BODY–SSC13 / TRIM–STS304 / SEAT–PTFE, FB, GEAR
VB	50A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY–SSC13 / TRIM–STS304 / SEAT–PTFE, FB, LEVER
VF	350A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY–GCD450 / TRIM–STS304 / EPDM–LINED, GEAR
F	350A	7EA	FLG SO, KS 10K, RF, STS304
F	300A	2EA	FLG SO, KS 10K, RF, STS304
F	50A	1EA	FLG SO, KS 10K, RF, STS304
FB	50A	1EA	FLG BLIND, KS 10K, RF, STS304
G	350A	8EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	300A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	50A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	350A	6SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	300A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	50A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
BL	350A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)
LOTO	350A	1EA	LOTO

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXH * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	CONTRACTOR SK ecoplant	PROJECT NAME M15X Ph4 Project
	THICK.	–	DESIGN	3.1	60	UNIT	4FA								
			OPERATING	2.0	AMB	REF. PLAN DWG. CJ–M15W14–PI–A30–040006–A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY WW–SLD	JOB NO. 22330D
						REF. P & ID CJ–M15W14–PR–A20–000115								LINE NO. 350A–WW–SLD–ST1–N–8577	SH'T NO. 01 / 01 REV. 0

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